



EVB-LAN7801-KSZ9031

Quick Start Guide

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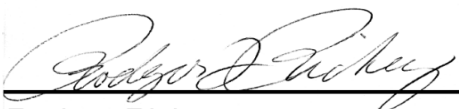
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Rodger Richey
Director of Development Tools



Date

NOTES:

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NOTES:

Preface

NOTICE TO CUSTOMERS

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Documents are identified with a “DS” number. This number is located on the bottom of each page, in front of the page number. The numbering convention for the DS number is “DSXXXXXA”, where “XXXXX” is the document number and “A” is the revision level of the document.

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INTRODUCTION

This chapter contains general information that will be useful to know before using and configuring the EVB-LAN7801-KSZ9031. Items discussed in this chapter include:

- [Document Layout](#)
- [Conventions Used in this Guide](#)
- [The Microchip Web Site](#)
- [Development Systems Customer Change Notification Service](#)
- [Customer Support](#)
- [Document Revision History](#)

DOCUMENT LAYOUT

This document describes how to use the EVB-LAN7801-KSZ9031. The manual layout is as follows:

- **Chapter 1. “Overview”** – Shows a brief description of the EVB-LAN7801-KSZ9031 board.
- **Chapter 2. “EVB-LAN7801-KSZ9031”** – Provides instructions about the board design and jumper configuration.
- **Appendix A. “EVB-LAN7801-KSZ9031 Evaluation Board Schematics”** – This appendix shows how to set up Master in Windows.

CONVENTIONS USED IN THIS GUIDE

This manual uses the following documentation conventions:

DOCUMENTATION CONVENTIONS

Description	Represents	Examples
Arial font:		
Italic characters	Referenced books	<i>MPLAB® IDE User's Guide</i>
	Emphasized text	...is the <i>only</i> compiler...
Initial caps	A window	the Output window
	A dialog	the Settings dialog
	A menu selection	select Enable Programmer
Quotes	A field name in a window or dialog	"Save project before build"
Underlined, italic text with right angle bracket	A menu path	<u><i>File>Save</i></u>
Bold characters	A dialog button	Click OK
	A tab	Click the Power tab
N'Rnnnn	A number in verilog format, where N is the total number of digits, R is the radix and n is a digit.	4'b0010, 2'hF1
Text in angle brackets < >	A key on the keyboard	Press <Enter>, <F1>
Courier New font:		
Plain Courier New	Sample source code	#define START
	Filenames	autoexec.bat
	File paths	c:\mcc18\h
	Keywords	_asm, _endasm, static
	Command-line options	-Opa+, -Opa-
	Bit values	0, 1
	Constants	0xFF, 'A'
Italic Courier New	A variable argument	<i>file.o</i> , where <i>file</i> can be any valid filename
Square brackets []	Optional arguments	mcc18 [options] <i>file</i> [options]
Curly brackets and pipe character: { }	Choice of mutually exclusive arguments; an OR selection	errorlevel {0 1}
Ellipses...	Replaces repeated text	var_name [, var_name...]
	Represents code supplied by user	void main (void) { ... }

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- **In-Circuit Debuggers** – The latest information on the Microchip in-circuit debuggers. This includes MPLAB ICD 3 in-circuit debuggers and PICkit 3 debug express.
- **MPLAB IDE** – The latest information on Microchip MPLAB IDE, the Windows Integrated Development Environment for development systems tools. This list is focused on the MPLAB IDE, MPLAB IDE Project Manager, MPLAB Editor and MPLAB SIM simulator, as well as general editing and debugging features.
- **Programmers** – The latest information on Microchip programmers. These include production programmers such as MPLAB REAL ICE in-circuit emulator, MPLAB ICD 3 in-circuit debugger and MPLAB PM3 device programmers. Also included are nonproduction development programmers such as PICSTART Plus and PIC-kit 2 and 3.

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- Technical Support

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Technical support is available through the web site at:

<http://www.microchip.com/support>

DOCUMENT REVISION HISTORY

Revisions	Section/Figure/Entry	Correction
DS50002605A (06-16-17)		Initial release of document

Chapter 1. Overview

1.1 INTRODUCTION

EVB-LAN7801-KSZ9031 is intended to enable evaluation LAN7801 with a known standard PHY device, KSZ9031RNX. Typically, customers' platforms will use a PHY other than standard Ethernet together with LAN7801.

EVB-LAN7801-KSZ9031 is intended to show LAN7801 in a known working environment ahead of more creative evaluation with alternative Physical interfaces.

LAN7801 is a USB 3.1 Gen 1 USB to Gigabit Ethernet bridge and KSZ9031RNX is a Gigabit Ethernet PHY.

EVB-LAN7801-KSZ9031 is designed to be an easy to use USB to Ethernet bridge that can be used in a system without an Ethernet port, for example, to connect to a network.

The evaluation board is bus-powered and uses a standard 3.0 type B USB and RJ45 connector to connect to a network.

1.1.1 References

The following documents should be referenced when using this quick start guide.

- *LAN7801 device datasheet*
- *KSZ9031RNX device datasheet*
- *EVB-LAN7801-KSZ9031 board's circuit schematic*
- *LAN7800/LAN7800i software user's guide*

NOTES:

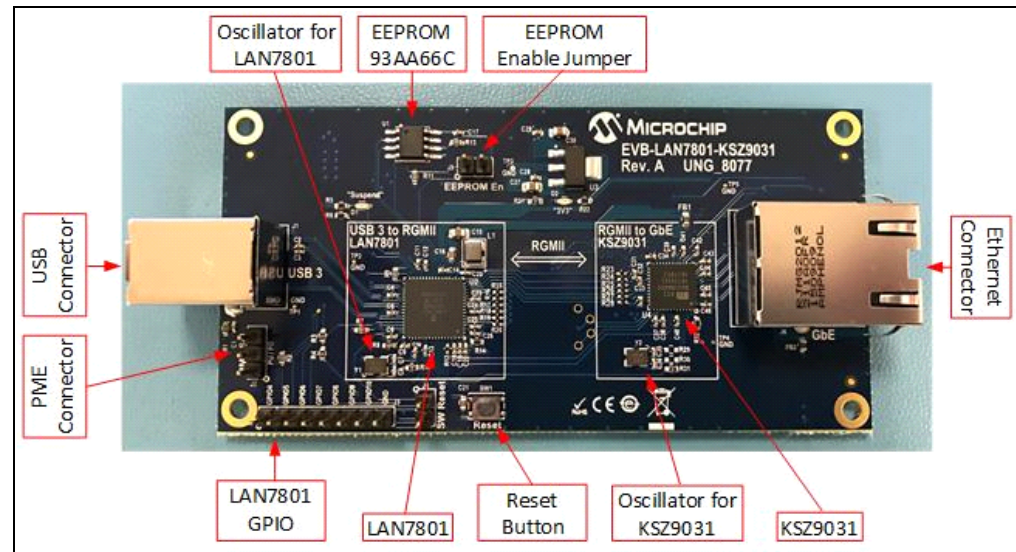
Chapter 2. EVB-LAN7801-KSZ9031

2.1 EVB-LAN7801-KSZ9031 BOARD DESIGN

This section is an overview of the EVB-LAN7801-KSZ9031 board design. The evaluation board allows a user to quickly evaluate LAN7801, as USB bridge device, with a known standard Ethernet PHY device, and KSZ9031RNX.

The LAN7801 bridge has an integrated Ethernet MAC interface. It can connect to an external MAC or a PHY. In this case, it is connected to a PHY, KSZ9031, to be used in the network. An Reduced Gigabit Media Independent RGMII (Interface) is utilized between the MAC (LAN7801) and PHY (KSZ9031). PHY configuration is achieved via the Serial Management Interface (SMI) between MAC & PHY.

FIGURE 2-1: EVB-LAN7801-KSZ9031 TOP VIEW



2.1.1 Power

EVB-LAN7801-KSZ9031 is a bus powered evaluation board. It receives power from the upstream host when it is connected via USB. There are two LDOs, MCP1826S and MIC5209, that supply regulated power to the LAN7801 and KSZ9031RNX. The MCP1826S is a 1A linear regulator that provides high-current and low-output voltage. It is a good choice for sub-1.8V output voltage applications that have high current demands. The MIC5209 is an efficient linear voltage regulator with very low dropout voltage, typically 10 mV at light loads and less than 500 mV at full load, with better than 1% output voltage accuracy.

2.1.2 LAN7801 GPIO Connector

J3 provides an easy access to GPIO4 through GPIO10, general purpose I/Os of the LAN7801 bridge. These GPIO pins are each fully programmable as a push-pull output, an open-drain output, or a Schmitt-triggered input. The GENERAL PURPOSE IO CONFIGURATION 1 (GPIO_CFG1) register is used to program the GPIO functionality. An external EEPROM is used to program the GPIO functionality.

This board uses the 93AA66C EEPROM to program the program GPIOs. The GPIO0 through GPIO3 pins on the LAN7801 device are multiplexed with the EEPROM controller.

2.1.3 EEPROM

An optional 93AA66C EEPROM is available to configure the LAN7801 device parameters. A jumper fitted at J5 enables this EEPROM. When the board is powered or a reset is issued and J5 is set, and a valid configuration is found with the EEPROM, LAN7801 will load parameters from the EEPROM before presenting itself for enumeration with the USB host.

The EEPROM can be accessed to manipulate the LAN7801 parameters using LAN78xx Windows Command Line Utilities. The software utility and its user's manual are available from the LAN7801 device's product page.

2.1.4 Power Management Event

EVB-LAN7801-KSZ9031 provides a mechanism for waking up a host system via the Power Management Event (PME) mode of operation. Connecting a jumper on pin one and two of **J2** enables the PME mode. Please refer to the device data sheet for the detailed information about the PME modes.

2.1.5 KSZ9031RNX Gigabit PHY

LAN7801 Ethernet MAC is connected to KSZ9031RNX PHY by RGMII interface. The default parameters of PHY are configured using the strap options of the device. For example, the MODE0 through MODE3 pins of the PHY have pull-up resistors. This means that the after reset this device advertises all capabilities such as, 10/100/1000 speed and half-/full-duplex.

Appendix A. EVB-LAN7801-KSZ9031 Evaluation Board Schematics

A.1 INTRODUCTION

This appendix shows the EVB-LAN7801-KSZ9031 Evaluation Board Schematics.

FIGURE A-1: USB LAN7801_1

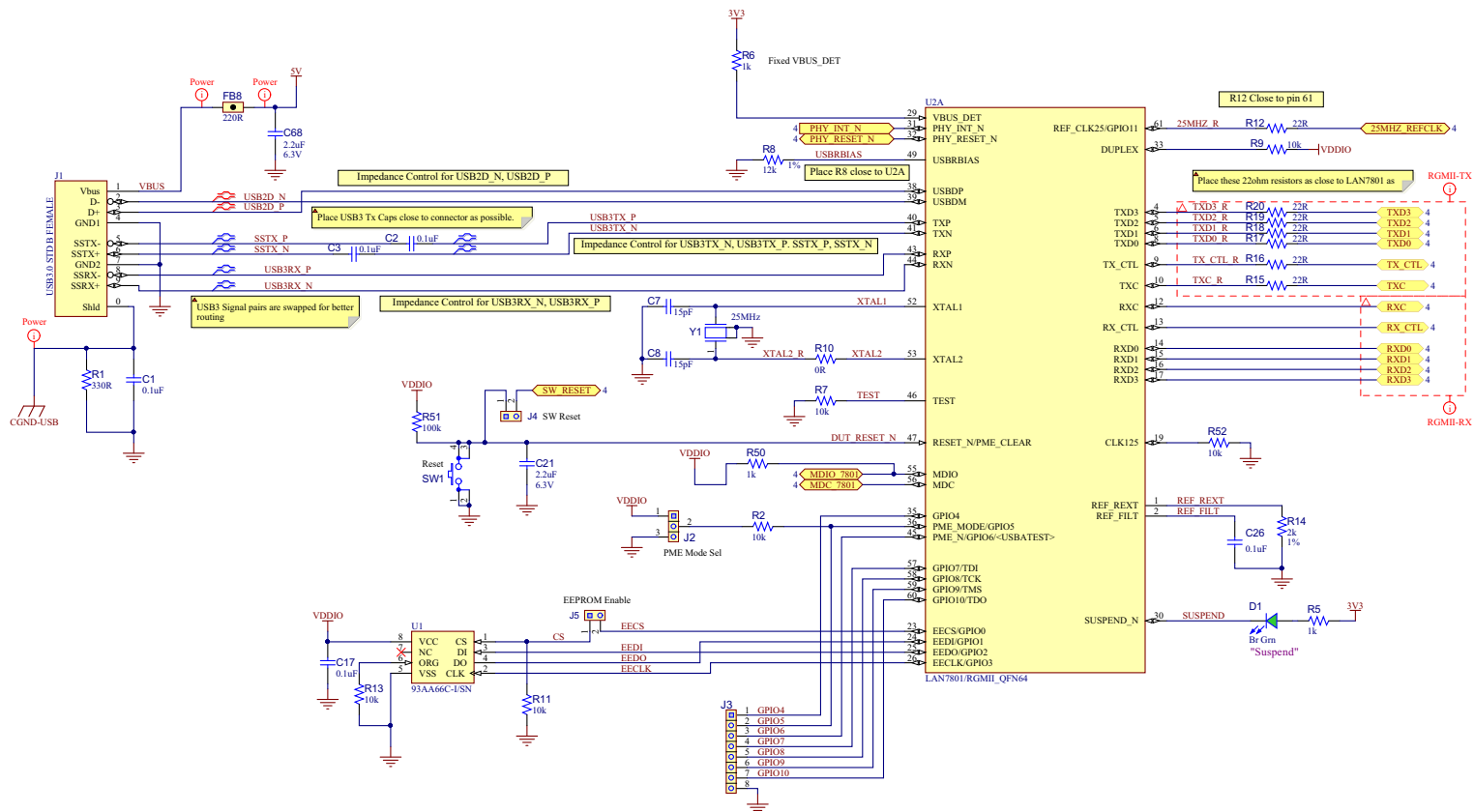
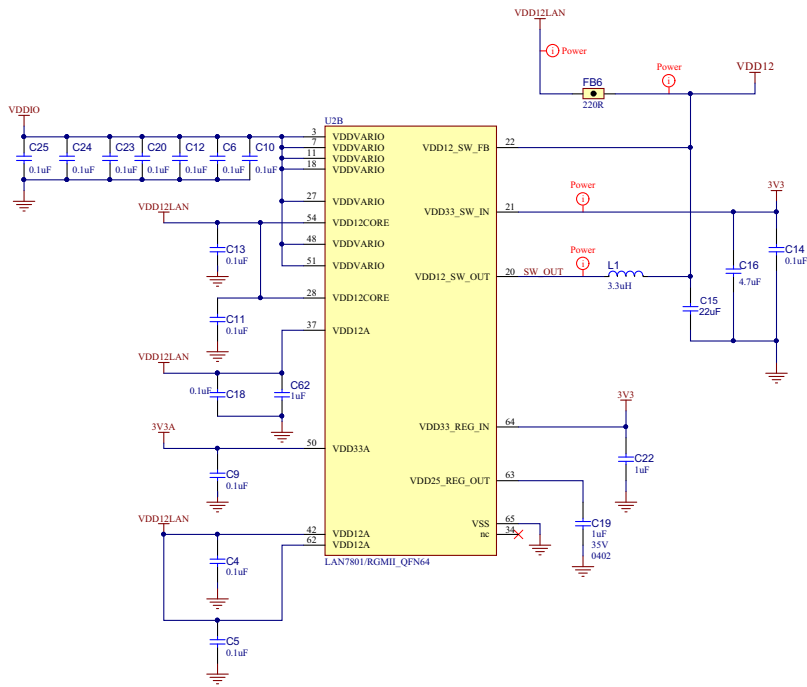
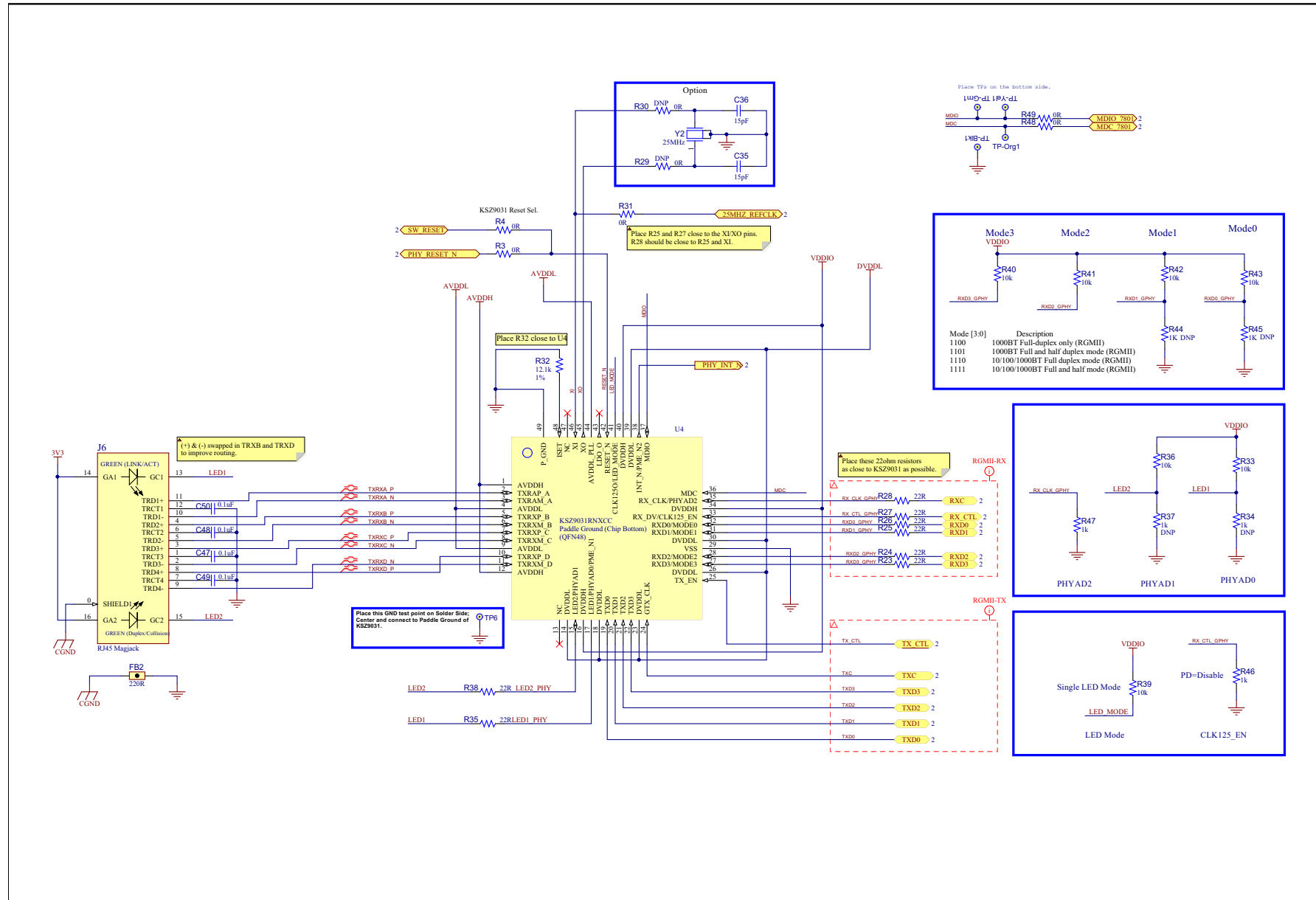


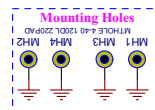
FIGURE A-2: USB LAN7801_PWR



EVB-LAN7801-KSZ9031 Quick Start Guide



POWER



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